Online Learning Platform – SQL Analysis Summary Report

Project Overview

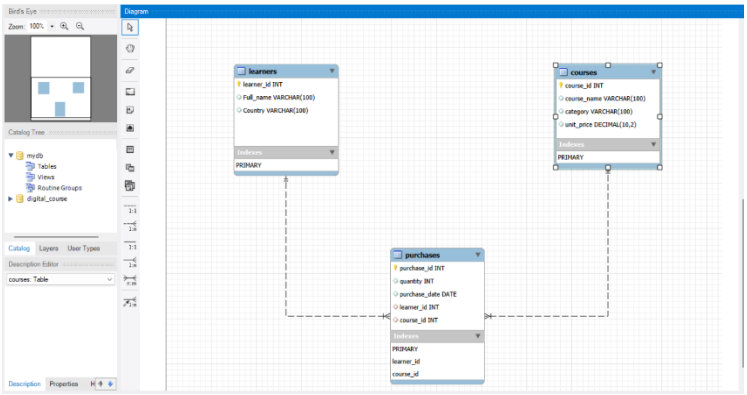
This project involved designing and analyzing a relational database for an online learning platform using MySQL. The database included three tables: Learners, Courses, and Purchases. SQL joins, aggregations, and analytical queries were used to evaluate learner behavior, course performance, and revenue generation.

The objective was to extract meaningful business insights using SQL techniques such as INNER JOIN, LEFT JOIN, RIGHT JOIN, GROUP BY, HAVING, ORDER BY, and derived calculations.

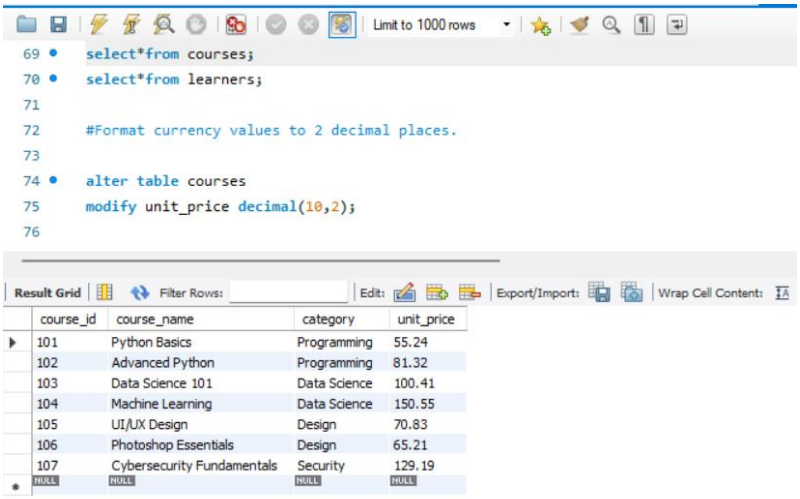
Problem Statement:

An online learning platform sells various digital courses to learners across different countries. Each learner can purchase multiple courses, and each course, belongs to a specific category. The management team wants to analyze purchase data to understand sales trends, learner behavior, and popular course categories. The task is to use MySQL to design and query a relational database that helps derive meaningful insights from this e-learning purchase data.

Relationship – ER Diagram:



Format currency values to 2 decimals



```
select*from courses;
select*from learners;

#Format currency values to 2 decimal places.

alter table courses
modify unit_price decimal(10,2);
```

Result Grid

course_id	course_name	category	unit_price
101	Python Basics	Programming	55.24
102	Advanced Python	Programming	81.32
103	Data Science 101	Data Science	100.41
104	Machine Learning	Data Science	150.55
105	UI/UX Design	Design	70.83
106	Photoshop Essentials	Design	65.21
107	Cybersecurity Fundamentals	Security	129.19
NULL	NULL	NULL	NULL

Q1: Display each learner's total spending (quantity × unit_price) along with their country.

```
Select (quantity*unit_price) total_amount, country
from purchases p
join learners l
On l.learner_id=p.learner_id
Join Courses c
On c.course_id=p.course_id;
```

total_amount	country
55.24	USA
100.41	USA
162.64	UK
70.83	UK
100.41	India
150.55	India
55.24	Canada
141.66	Canada
65.21	USA
803.28	Canada
569.24	UK

Q2: Find the top 3 most purchased courses based on total quantity sold.

```
Select course_name, sum(Quantity) as Total_Quantity
from purchases p
join courses c
On c.course_id=p.course_id
group by c.course_name
order by total_quantity desc
limit 3;
```

course_name	Total_Quantity
Data Science 101	10
Advanced Python	9
UI/UX Design	3

Q3: Show each course category's total revenue and the number of unique learners who purchased from that category.

```
#Q3. Show each course category's total revenue and the number of unique learners who purchased from that category.
select Category, sum(unit_price*Quantity) Total_revenue, count(distinct p.learner_id) Unique_Purchase
from Purchases p
join courses c
On c.course_id =p.course_id
group by category
order by Total_revenue desc;
```

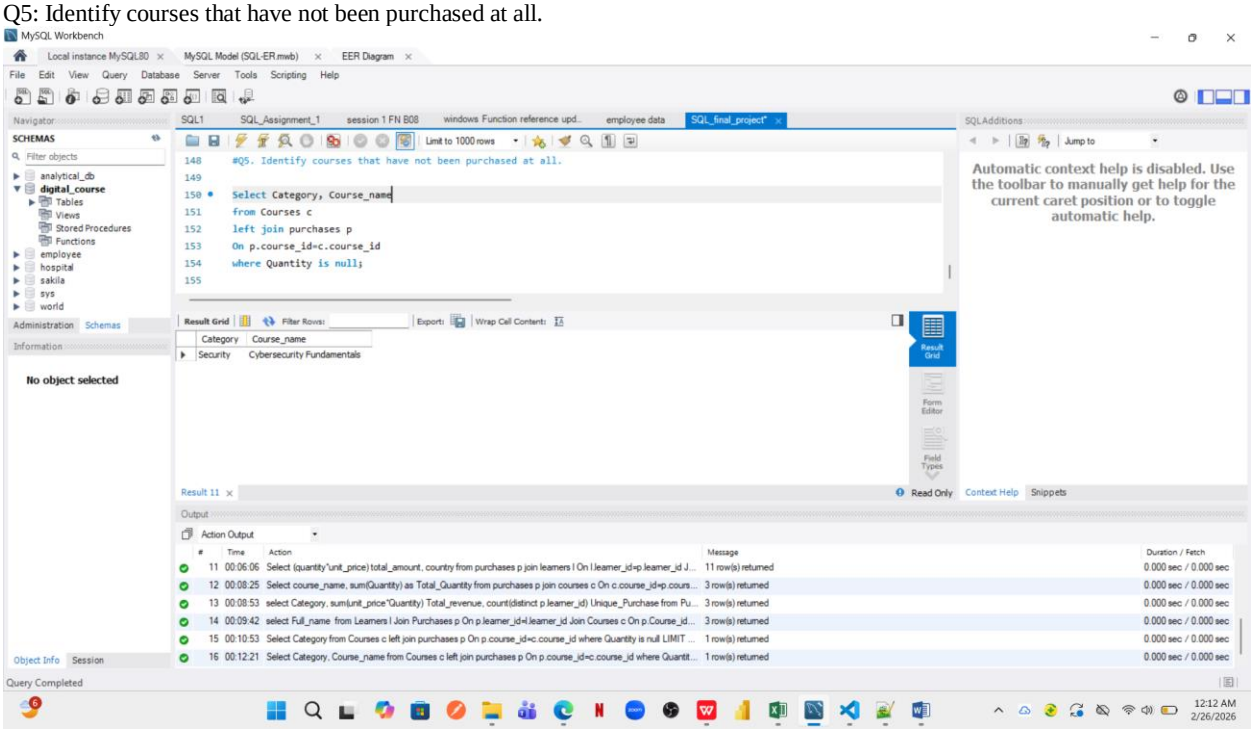
Category	Total_revenue	Unique_Purchase
Data Science	1154.65	3
Programming	842.36	3
Design	277.70	3

Q4: List all learners who have purchased courses from more than one category

```
select Full_name
from Learners l
Join Purchases p
On p.learner_id=l.learner_id
Join Courses c
On p.Course_id= c.Course_id
group by l.learner_id, full_name
having count(distinct category)>1;
```

Full_name
Alice Johnson
Bob Smith
Diana Patel

Q5: Identify courses that have not been purchased at all.



Key Insights from Analysis

◆ Learner Spending Patterns

- Total spending varied significantly among learners.
- Learners who purchased multiple courses or higher-priced courses contributed the most revenue.
- Some learners purchased from multiple categories, indicating broader interests and higher engagement.

◆ Top Performing Courses

- Programming-related courses recorded the highest total quantity sold.
- High-demand courses directly influenced overall platform revenue.
- Courses with repeat purchases showed strong learner preference.

◆ Category-Level Revenue

- Programming and Data Science categories generated the highest revenue.
- Categories with more unique learners indicate stronger market demand.
- Some categories showed lower engagement compared to others.

◆ Cross-Category Purchasing Behavior

- Certain learners purchased from more than one category.
- These learners represent high-value customers who are open to diverse learning paths.

◆ Underperforming Courses

- At least one course had no purchases.
- This indicates possible issues in marketing visibility, pricing strategy, or demand.

Business Recommendations

1. **Focus marketing efforts on top-performing categories** such as Programming and Data Science to maximize revenue growth.
2. **Introduce bundled course offers** to encourage cross-category purchases and increase average order value.
3. **Target high-spending learners** with loyalty programs, discounts, or early access to new courses.
4. **Reassess underperforming courses** by reviewing pricing, improving course descriptions, or running promotional campaigns.
5. Use data-driven insights regularly to monitor trends and adjust business strategy accordingly.

Skills Demonstrated

- Database creation and schema design
- Data insertion and relationship management
- Multi-table joins (INNER, LEFT, RIGHT)
- Aggregation and filtering using GROUP BY and HAVING
- Revenue calculations and derived columns
- Business-oriented SQL analysis
- Clear result formatting and reporting

Conclusion

The SQL-based analysis provided clear insights into learner behavior, course performance, and revenue distribution. By identifying top-performing categories, high-value learners, and underperforming courses, the platform can make informed, data-driven decisions. Continuous monitoring of purchase trends and category performance will support strategic growth and improve overall business performance.